Journal of Personality and Social Psychology

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Online First Publication, April 24, 2025. https://dx.doi.org/10.1037/pspp0000557

AMERICAN PSYCHOLOGICAL ASSOCIATION

Journal of Personality and Social Psychology: Personality Processes and Individual Differences

ISSN: 0022-3514

https://doi.org/10.1037/pspp0000557

Comprehensive Personality Structure in the Persian Language: High-Dimensionality Analyses of Trait Adjectives

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It is widely known that all languages have personality-trait concepts, but more controversial is how these concepts are organized (structured) based on application to actual human targets of description. Many assume that Big Five factors provide a universally applicable structural template, but evidence beyond European languages has particularly undermined this premise. The comparative reproducibility, across cultures, of structures of few broad factors (more parsimonious) versus many fine-grained factors (more comprehensive and predictive) also remains unclear. Here, issues of reproducibility and universality are examined in reference to real-person data from the non-Western cultural context associated with the Persian language. Self-reports from 767 Iranian adults employing 360 Persian personality terms were analyzed by both high-dimensionality and traditional approaches. Imported Big Five and six-factor structures were not well-supported. Although few-factor structures of one or eight factors showed the highest reproducibility (across method variations) in absolute terms, more comprehensive structures (of 20 or 42 factors) accounting for far more variance were nearly as reproducible, suggesting distinct structural layers, hierarchical only to a partial degree in a hierarchy. The 42-factor structure revealed not just many contents quite comparable to well-understood Western trait concepts but also some intriguing relatively culture-specific contents, thus indicating in a novel way how personality-structure studies can be revealing of culture and cultural differences.

Keywords: Big Five, factor analysis, lexical studies, HEXACO, culture

Basic research on the structure of personality attributes aims to map the most important dimensions in personality. Such dimensions organize the basic lines along which individuals differ in their tendencies regarding behavior, affect, and thinking. A productive approach for identifying these basic dimensions has been relying on the representation of personality variables in dictionary lexicons of various languages, based on a lexical rationale. By this rationale, the most important individual differences tend to become sedimented enduringly into a finite set of terms in natural language, and the most salient lexical terms—arguably representative of the full population of terms—are converted to a questionnaire format for ratings of actual people. Such lexical studies provide a relatively unbiased perspective on what might be an optimal structure for the personality attributes of human targets (even if downstream from there phrased items representing an optimal structure are more advantageous). The approach is fundamentally emic, identifying structure in a single language and cultural context but adds an etic aspect by comparing various emic-study results to develop inferences regarding potential universal dimensions.[1]

To ensure sufficient applicability to populations around the world—not just Western populations—it is crucial to conduct these studies in prominent non-Western cultural contexts (Thalmayer, Toscanelli, & Arnett, 2021). Indeed, some challenges to established Western-based models have arisen based on personality-structure results in non-Western contexts (Cheung et al., 2001; Thalmayer et al., 2020; Thalmayer, Job, et al., 2021; Zhou et al., 2009). Middle

Joanne M. Chung served as action editor.
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Preregistered analysis plans associated with this study and all code, syntax, and research materials not included in the article are publicly available on the Open Science Framework at https://osf.io/ydbgt/?view_only=7cdc709d395f49aaa878c5e9216ec402. The project was aided by input from Reihaneh Aslanzadeh and Lewis R. Goldberg.

Naser Aghababaei played a lead role in funding acquisition, investigation, variable selection, and data collection, a supporting role in writing–review and editing, and an equal role in conceptualization, data curation, methodology, project administration, and software. Gerard Saucier played a lead role in supervision and writing–original draft, a supporting role in investigation and project administration, and an equal role in conceptualization, data curation, formal analysis, methodology, and software. Vinita Vader played a supporting role in data curation and writing–review and editing and an equal role in formal analysis, methodology, and software. Pooya Razavi played a lead role in judgments of factor interpretability, a supporting role in writing–review and editing, and an equal role in conceptualization.

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[1] Another approach frequently labeled as a combined emic–etic approach to personality (e.g., Cheung et al., 2001) understands emic and etic slightly differently. There, on the emic side, one finds less emphasis on the lexicon as sedimented in dictionaries and more on folk representations of traits as evident in wordings and phrasings found in literary sources or in free-response descriptions. On the etic side, it emphasizes comparison to established models developed in Western contexts (e.g., the Big Five). This approach is also more oriented toward near-term development and validation of assessment instruments. By contrast, the lexical approach is more focused on examining single words and on answering theory-imbued questions like “what is the ideal structure for this one particular language” and “what structures are most universal across languages”; there is less deference to established questionnaire models and less overt focus on the pragmatics of downstream questionnaire development.

Eastern contexts are particularly poorly represented in contemporary psychological studies (Krys et al., 2025). Such a context is the focus of the present study—a lexical study of personality in the Persian language.

The Persian Lexicon of Personality Description

The Persian (Farsi) language is a primary language in Iran and also prominent in two other countries. Persian in Iran is mutually intelligible with Dari Persian in Afghanistan and with Tajiki Persian in Tajikistan and Uzbekistan. Overall, Persian is the first language of over 70 million people. The number of speakers alone underlines the importance of this context.

The Persian language has some common roots with English because both English and Persian are Indo-European languages. That is, Persian, English, and many other European and Asian languages have a common vocabulary core from a reconstructed ancestor language named Proto-Indo-European, existing over 5,000 years ago. On the other hand, Persian has been heavily influenced by Arabic, one of the non-Proto-Indo-European-derived languages (from the Semitic branch of the Afro-Asiatic language family). Although there are a surprising number of Arabic-derived words in English as well, the impact of Arabic on Persian vocabulary is of much greater magnitude. Thus, while we might expect Persian lexical personality structure to have a few affinities with many European structures based on ancient linguistic linkages, the effects of Arabic on Persian as well as the effects of a quite distinct cultural and religious context would lead us to expect that the Persian pattern for classifying personality traits may be quite different.

An earlier study of personality description in the Persian language drew on the selection of adjectives in a project, with the eventual work published by Farahani et al. (2016). In the initial stage, four postgraduate students of Persian literature extracted 11,530 adjectives from several Persian dictionaries and books (including 20 novels and one magazine). By successive stages, a broader team reduced the list to some 6,500 adjectives and then to 5,600. A trio of psychology faculty members reduced the list to 1,360 terms that were judged to have personality-descriptive relevance, which were then reduced to 544 based on judgments of assessed familiarity of the term, and then to 207 judged useful for description of stable traits. Finally, among the 207 dispositional trait adjectives, "words that had strong negative value or that could be seen as breaching social norms or values" were removed, leading to a final list of 126 terms. These were administered to 1,200 male and 1,200 female high school students in Tehran, Iran; 2,279 forms were useable in analyses. Structures of two to six factors were examined; there were some loose correspondences (correlations in the .40–.65 range) with scales based on 71 (of what were judged to be) Big Five marker items selected from among the 126 items.

There are several crucial limitations to Farahani et al.'s (2016) study. Although the sample of participants was substantial in size, the sample of adjectives was small, too small to facilitate extensive exploration of structure beyond just a few factors. The heavy reliance on judgments of faculty members—from departments of psychology and of Persian literature and language—opens the possibility that their final list of adjectives may be the result of prestructuring or "expert bias" (Church et al., 1998; Saucier & Goldberg, 1996) in the selection of terms. Moreover, typically, lexical-study projects have concentrated on a single comprehensive dictionary to provide a representative sample of a language's person-descriptive adjectives. Farahani et al. relied on more than one dictionary, which is probably unnecessary and does somewhat complicate the repeatability of the research, unless there is a clear sequencing in the use of the dictionaries or strict rules are applied to how one versus the other dictionary is used. More unconventional (though not unheard of; De Raad & Barelds, 2008) was their procedure of extracting terms also from literary sources such as novels; this choice weakens the argument that one has a representative selection of person descriptors because questions can be raised about how fairly a set of literary texts represent the language and about biases accruing from the nature of the (inevitably) small number of texts selected. Perhaps even more concerning, Farahani et al. also apparently omitted many terms on the basis that they were negative, unappealing, or unwanted traits (e.g., one translatable as "ill mannered")—apparently on the advice of their three psychology experts. Elimination of extremely evaluative terms (e.g., evil) has not been unusual in lexical-study procedures, but in the research of Farahani et al., it may have cut well into descriptors (representing the "dark" side of personality) that are not the most extreme in their evaluation.

Work by Farahani et al. (2016) is commendable as the first attempt to conduct a lexical study on Persian language. However, one might expect quite different results if a more standard systematic lexical study of Persian-language descriptors was conducted with a much larger sample of trait descriptors. This would facilitate adding on a high-dimensionality analytic paradigm. Results of a factor analysis do depend on the input. If types of descriptive content are entirely or even mostly excluded—as will certainly happen with the selection of only 126 terms—the resulting structure will be truncated and perhaps distorted. A relevant (and thorough) study of modern standard Arabic by Zeinoun et al. (2018) has the same limitation, as it arrived at only 167 descriptors, which proved (as for Farahani et al., 2016) useful only for identifying a relatively small number of factors (in that case, seven). Real-person data rather than internet text-scraping data are essential to this end, inasmuch as methods using text data (e.g., Cutler & Condon, 2023) have heretofore failed to support more than a very few broad factors, whereas lexical studies administering questionnaires to real persons clearly demonstrate no such limitation. Because the high-dimensionality analytic paradigm is not familiar to all in the field, we review it in some detail, next.

The High-Dimensionality Perspective on Personality Structure

Past applications of a lexical approach in over two dozen languages (mostly of European origin) have emphasized relatively parsimonious structures with six or fewer dimensions. Structures at this broad level have several advantages. The small set of dimensions is easy to remember and can be measured with a brief set of items.

Limitations of Big Five Structures

On the other hand, there are severe limitations associated with sole reliance on such simplified structures of personality variation. These broad factors are essentially collages of diverse content (i.e., divergent facets). Because of this collagelike nature, considerable fuzziness inheres in their interpretation; moreover, there are difficulties comparing the "collage" generated in one language (or even two data sets in one language) when, as often happens, one part of the collage is reproduced, but not another. Thus, there is difficulty in *decisively* replicating popular broad structures (such as the Big Five) across very different cultural-linguistic contexts.

Broad factors also compromise content coverage of the personality domain and predictive capacity; lexical five-factor structures typically only account for roughly 25% of the variance in the intercorrelations of the variables; when more factors are introduced, the predictive capacity tends to increase roughly as the percent of variance accounted for (i.e., the comprehensiveness) increases (Saucier & Iurino, 2020). An increase in prediction could be expected, inasmuch as when more specific unique sources of variance are captured by the predictors, the higher will tend to be the variance accounted for by the predictors (Mershon & Gorsuch, 1988).

A more thorough attempt to understand personality-trait structure will examine not only the low-hanging fruit of broad factors but also harder-to-reach complex structures that may be equally robust (replicable) and ultimately superior for predictive purposes. Recent studies delving further into high-dimensionality structure (Saucier et al., 2020, Study 1; Saucier & Iurino, 2020; Thalmayer et al., 2020; Thalmayer, Job, et al., 2021; Vader & Saucier, 2025) make it clear that one can readily find more than just five or six useful independent and relatively robust dimensions of personality in a language.

A widely valued goal among researchers is the identification of a consensual structure that is not only robust within any cultural-linguistic context but also between these contexts: a universal structural model for personality. Proposed structures of five and six factors/dimensions have aspired to this goal but run up against replicability challenges in many contexts (especially non-Western ones, e.g., Cheung et al., 2001). With regard to the Big Five structure, the clearest support has been found in Germanic (e.g., De Raad et al., 1992) and Slavic (e.g., Mlačić & Ostendorf, 2005) languages, conducing to the generalization that this is an emic structure especially for north-European languages. The six-factor structure has seen better support in some other European languages and in a few beyond but weak replication elsewhere (e.g., Thalmayer et al., 2020; Thalmayer, Job, et al., 2021). There is more support for three potentially universal dimensions (De Raad et al., 2010, 2014) that are largely three out of the Big Five and perhaps even more support for two such dimensions (De Raad et al., 2014; Saucier, Thalmayer, & Bel-Bahar, 2014; see also Zeinoun et al., 2018) that are labeled as the Big Two, with independent dimensions of social self-regulation and dynamism. The nature of a potentially robust one-dimensional structure—emphasizing good/bad evaluation—has attracted less attention, although studies of moral perception (e.g., Goodwin et al., 2014) highlight just such a dimension and classic studies of dimensions of meaning as applied to nonhuman objects found a large first (unrotated) factor of a broad "evaluation" interpretation (Osgood et al., 1975; Osgood & Suci, 1955).

What Makes a Good Candidate Structure?

It may be unlikely that as many as, say, 20 dimensions (as in Saucier & Iurino, 2020) will find perfect matches across languages. However, the identification of granular, more unambiguously interpretable factors/dimensions affords a different paradigm for cross-cultural comparisons, one more analogous to that used by emotion researchers seeking to determine which granular emotion concepts (e.g., fear, anger, guilt) are more versus less universal.

This different paradigm, yielding insights harder to gain with collagelike broad factors, might go as follows: One generates a large number of robust dimensions (e.g., 20, or 40) in each language, comparing multiple languages to determine which dimensions emerge most ubiquitously and which, in contrast, are the most culture specific. In the domain of emotion, researchers have identified a widely referenced set of six "basic emotions" by taking into account a much larger number of granular emotion concepts, finding these six granular concepts as the ones most salient across most contexts (by criteria applicable to emotion). However, far more than six granular emotions can be identified (Cowen & Keltner, 2017), and apparently this would be so within most any cultural context.

Replicability of structural models of personality can rely on multiple criteria: not just the across-rotation-method criterion impressively applied by Goldberg (1990). Factors that meet the baseline criteria—being sufficiently sized and meaningfully interpretable—can be examined for resiliency across variant methods, such as not only differing rotation algorithms but also the use of self- versus peer ratings, or of original versus ipsatized data, and reproducibility across meaningful subpopulations within the sample (e.g., women vs. other gender identities).

Methodological Considerations

Essential to the high-dimensionality approach to personality structure (originating with Saucier & Iurino, 2020) is the generation of multiple candidate-structure models, each model standing out for its comprehensiveness and the interpretability of all its constituent factors. The candidate structures are then compared with respect to their robustness—their replicability across a set of method variations. The present study generally follows previously developed standard procedures for creating these comparisons. Here, we review six criteria to be met in defining a viable candidate structural model; all of these have been used in one or more other previous studies before this study of the Persian lexicon.

1. *Represent previously emphasized broad levels of structure*. For purposes of comparison, few-factor structural models (of the sort almost exclusively emphasized in lexical studies before 2020) are always included. This would include structures of six, two, three, five, and six factors as well as, for completeness, four-factor structures. Following the method norms most commonly employed since the 1990s, these structures preferentially draw on self-ratings, ipsatized data, and principal components with orthogonal varimax rotation procedures.[2]

[2] Principal component analysis has been favored in lexical studies (a) because these studies tend to heavily used factor scores and principal component analysis generates exact factor scores, whereas other factor methods only (by one of a set of not universally accepted methods) generates estimates of such; (b) because of the large number of variables analyzed, these studies sometimes use samples or subsamples wherein the number of variables exceeds that of subjects, which PCA rather uniquely can handle; and (c) because previous studies used PCA, comparability is enhanced by continuing to do so. Lexical studies identifying emic structures employ exploratory factor-analytic rotation procedures and not confirmatory rotation approaches such as, for example, Procrustes. Procrustes rotation requires that one already has a set of target coefficients for factors and that prior to that one already knows how many factors there are. This is not appropriate for high-dimensionality studies of personality-attribute structure, like this one, wherein one does not know in advance how many factors or what the exact content of the factors will be.

2. *Employ a variety of data sources and factor rotations*. Beyond those few-factor levels, a set of high-dimensionality structures is identified using parallel analysis (PA; Horn, 1965) to determine a ceiling on the number of factors. Here, in identifying candidate structures, no preference is given to ipsatized over original (raw) data for analysis. Several widely available rotation methods are applied. In addition to varimax rotation, an oblique-rotation method (oblimin) unrelated to varimax is applied, along with an orthogonal rotation more capable of yielding a large number of meaningful factors (equamax). A candidate structure is identified for each combination of data type (ipsatized or original data) and the three rotation methods; this defines six "lanes" in which a structural model is identified for comparison. This approach tends to minimize the degree to which results are biased by arbitrary method choices (e.g., whether to ipsatize, which rotation method to employ).

3. *Identify a variegated set of plausible high-dimensionality structures*. There would be little interest in factors defined by the unique (possibly error) variance of single variables or in representing accidental combinations of unrelated concepts (which might arise not from zero-order correlations but from correlated residuals). The candidate structure selected in each lane is that structure defined as the one with the largest number of factors wherein (a) all factors are sufficiently sized and (b) all factors are found to be interpretable. Because some factor solutions include one or more factors that are insufficiently sized and/or uninterpretable, the search for a candidate structure proceeds iteratively downward from the maximum that is set by PA, until a solution meeting all three criteria is identified. "Sufficient size" has previously been defined based on a fraction of the total number of variables, whereby the minimum number of salient variables (terms having their highest loading on the factor) is a number near the threshold of $k/200$, where k is the total number of terms (variables) analyzed. Rationally, this number should always be at least 2, so as to avoid an overlong string of singleton factors wherein it is difficult to apply an interpretability test. The salient loadings must be of fairly substantial size (in the vicinity of .40 magnitude or higher). Interpretability means that a researcher (at least one investigator) is demonstrably able to find the factor interpretable (i.e., it is not consensually judged to be uninterpretable).

4. *Set alternative ceilings on the number of factors*. It could be erroneous to assume that the ceiling set by PA is an absolute limit. PA computations are based on *unrotated* factors, which characteristically become quite small after the initial factors are generated, although with factor rotation most of those quite-small unrotated factors become larger. Moreover, PA discounts a factor when it is too small (i.e., smaller than what random data would tend to generate), but this may bias it against very high-dimensionality structure wherein useful factors may not be larger in size than what random data generates. For example, Vader and Saucier (2025) have found a relatively robust 32-factor structure in English nouns that was well above the PA recommendation. So as not to fall prey to erroneous assumptions, it is useful to compare the PA limit with some alternative. One might simply use PA as a starting point and proceed iteratively upward rather than downward (the approach applied here).[3] Because in recent studies varimax rotations have never yielded a suitable structure with as many factors as PA indicates, this procedure is dropped for this very-high-dimensionality tier of structure, and rotation methods more suitable for the very-high-dimensionality task (oblimin and equamax) are retained, with again separate models for ipsatized and for original data.

5. *Identify which structure is the most robust (i.e., replicable across method variations)*. The candidate structures, consisting of the six few-factor models defined under Item 1 above, plus six high-dimensionality structural candidates defined by relying on the PA ceiling, and then four additional very-high-dimensionality candidates defined by discarding the PA ceiling, are compared with respect to replicability across method variations. One criterion is robustness across rotation method, when an orthogonal and an oblique rotation are compared (using correlations between factor scores). Another criterion is robustness across data type, oriented to ensure that the structure is not artifactually dependent on either the ipsatization procedure or the lack thereof; this comparison involves determining the proportion of variance shared between sets via canonical-correlation procedures. A third criterion would be, ideally, congruence of structure between self- and peer ratings, but if no peer ratings are available, one can examine instead congruence of structure across two demographic subsets of the sample to ensure that structure does not overly depend on any demographic subpopulation. Of great interest is which structure (among the 16 delineated above) generates the highest absolute robustness (averaged across the three criteria). However, because structures with many factors are likely to have more predictive capacity than those with few factors, one can expect a trade-off between absolute robustness and comprehensiveness. Therefore, also of great interest is which, among the more comprehensive models, stands out from the others for its relative robustness, that is, which of the more comprehensive models least compromises robustness.

[3] Alternatively, one could start with the number of factors that (as per Merenda, 1997) manages to account for 50% of the variance in the intercorrelations (approximately double what the Big Five accounts for) and proceed iteratively upward (the approach applied by Vader & Saucier, 2025) or apply the 60%-of-variance criterion that has been suggested as a number-of-factor guide by some psychometricians (Hair et al., 2010), using that as a ceiling and proceeding iteratively downward until all criteria are met (i.e., the largest number of factors that has entirely sufficiently sized and interpretable factors). Presumably, these three alternative approaches could lead to the same result; which approach is the most efficient should eventually be evaluated. Of note, Peterson's (2000) meta-analysis of factor analyses from 568 studies found that the average variance explained by a factor structure was 56.6%; only 30% of studies presented factor analysis structures accounting for less than 50% of variance, with less than 1/10 of studies presenting structures (like, typically, five factors in lexical studies) accounting for less than 30%. In this regard, the method in most lexical studies of personality—concentrating on just a few factors—is an anomaly. As for the minimum average partial (MAP) procedure of Velicer (1976), Saucier and Iurino (2020) found it less useful than PA for large numbers of single terms, and they came not to recommend it for that purpose.

> It has been typical that these comparisons have yielded at least two relatively robust structural models that have quite different numbers of factors, and these can be taken to indicate two (or more) strata identified in personality-attribute structure in the given language community. Unlike the now oft-employed "bass-ackwards" approach (Goldberg, 2006) for defining levels of structure, this method discards levels of structure that are relatively less robust and emphasizes only the levels found to be the most robust.

Based on the above considerations, and following the method outlined above, the present study derives personality dimensions based on the Persian lexicon. In addition to the broad (parsimonious) dimensionality structure, it also identifies the high-dimensionality structure of adjective personality descriptors in the Persian language. The present study is designed to go beyond various limitations of previous work and enable a high-dimensionality perspective on personality variation in a non-Western context.

Method

Selection of Terms

In this study, a contemporary Persian dictionary (Sadri Afshar, 2014) comprising about 70,000 entries was examined, independently, by the first author and an assistant (a teacher of sociology with a PhD degree who is fluent in Persian). In total, there were 2,996 person-descriptive adjectives initially selected.

In the second step, 11 judges used a popular procedure (described next, following Angleitner et al., 1990) to categorize these adjectives into one of 15 categories. A given word was classified as belonging to a particular category if it was indicated as such by at least seven judges.

The words were rated on the following 3-point scale: 1 = *the meaning of the word is not clear enough for me to complete subsequent ratings*, 2 = *the meaning of the word became clear to me only after giving it some thought*, 3 = *the meaning of the word is fairly clear to me*. If the meaning was clear enough (3), the judge moved to the next step, in which he or she had to decide whether the adjective could be imagined being used for the description of an individual or for the description of an individual's experience, behavior, or appearance. The words were rated on a 3-point scale. If the judge responded 1 (*impossible to imagine*) or 2 (*unusual; possible to imagine only under certain conditions*), the term was considered not personality relevant. Each adjective rated by the judge as person descriptive (easy to imagine as a person descriptor) was then supposed to be classified into one of 13 subcategories, which in turn fall into five superordinate categories: (1) dispositions, (2) temporary conditions, (3) social and reputational aspects, (4) overt characteristics and appearance, and (5) terms of limited utility. Category 1 was divided into two subcategories: Subcategory 1a was reserved for temperament and character traits and Subcategory 1b for abilities and talents or their absence. Category 2 contained three subcategories: (2a) experiential states (emotions, moods, and cognitions), (2b) physical and bodily states, and (2c) observable activities. Category 3 consisted of four subcategories: (3a) roles and relationships, (3b) social effects (reactions of others), (3c) pure evaluations, and (3d) attitudes and world views. Category 4 was divided into two subcategories: (4a) bodily characteristics pertaining to anatomy, constitution, and morphology and (4b) sociocultural aspects of appearance (looks and deportment). The fifth category consisted of two subcategories: (5a) context-specific and technical terms and (5b) metaphorical, vague, or outmoded terms. A sixth category was reserved for classifiable words: (6a) words the judge was not sure which category they belong to and (6b) words that they see not belonging to the abovementioned categories.

In the end, 432 adjectives were categorized as dispositions. Of the 432 words classified as dispositions, 72 were omitted based on their falling into any of the following exclusion categories: (a) adjectives of similar meaning with a word root redundant with another adjective, (b) swear words, (c) being specific to one gender, and (d) words that were related to intelligence (or lack thereof) rather than personality. Noteworthy that emotional states, attractiveness, or attitudes (e.g., religiousness) were not included because these were classified into independent nondisposition categories. The final list comprised 360 adjectives. A list of the 360 terms (in Persian, with two sets of English translations) can be found at https://osf.io/ydbgt/?view_only=7cdc709d395f49aaa878c5e9216ec402.

Participants

Self-ratings were gathered from an online sample using Google Forms. Data were collected during the course of the COVID-19 pandemic, when paper-and-pencil questionnaires were not feasible. Participants were recruited by circulating notification of the study's availability through the social-network groups of the first author and his colleagues (faculty professors) and as described below included many individuals with graduate (postbaccalaureate) education. The aim of the study was stated at the outset of the materials, and only those who explicitly indicated a willingness to participate could proceed to study participation. Participation was voluntary and without compensation. The study was approved at the SAMT Department of Behavioral Sciences.

Data collection proceeded till a large (N > 800) sample was achieved. Upon initial observation, 805 cases were found in the data; however, one case was eliminated due to excessive use of the same response (for over one third of the questions). Then, examination of the data revealed that 37 cases were duplicates (the same data from a respondent being mistakenly registered again). After removing these cases, there were 767 participants. Based on responses to basic demographic questions, the participants included 155 males and 612 females (15% and 85%, respectively), with an average participant age of 34.12 years. Overall, this tended to be a well-educated sample: Five participants (1%) had less than a full high school education, 54 (7%) had a high school education only, 17 (2%) had an associate degree, 258 (34%) had education at the bachelor's degree level, 354 (46%) had master's-level education, and 79 (10%) had doctorates; in other words, over half the sample had graduate degrees.

Materials

Materials were administered via Google Forms, with items administered in fixed order, the same for all respondents. Responses to each adjective were made using a 1 (*totally untrue*) to 5 (*totally true*) Likert scale, with the intervening values labeled as *somewhat untrue* (2), *neither true nor untrue* (3), and *somewhat true* (4). Data were ipsatized by standardizing each row (case) in the data, across the 360 adjectives, in accord with a procedure typically followed in lexical studies. However, original (raw) and ipsatized data were subsequently analyzed.

Transparency and Openness

The data-analysis procedures were specified in an analysis plan preregistered on the Open Science Framework at https://osf.io/ydbgt/?view_only=7cdc709d395f49aaa878c5e9216ec402. The analyses underlying the text found in Footnotes 3, 4, and 5 were not in the preregistration; these are not central analyses in a study like this and addressed issues not foreseen in advance. Footnotes 3 and 4 address issues that came up in the earlier round of this review process, whereas Footnote 5 involves post hoc analyses based on unanticipated patterns in results (four hierarchical tiers of structure) that could not be anticipated before the study was carried out. In addition, the manner in which results are presented in tables and graphs was not foreseen in the preregistration.

Data-analysis procedures are described below in the precise order found in the preregistration. Data were analyzed using R wherever feasible and with SPSS (Version 27) where suitable R-package routines could not be found. Code (or syntax) and data are publicly available on the Open Science Framework at https://osf.io/ydbgt/?view_only=7cdc709d395f49aaa878c5e9216ec402.

Analyses

Analysis procedures, specified in the analysis plan, followed a predetermined sequence of nine steps. The steps, combined so as to make a simplified set of five stages, are presented in Figure 1. The reasoning informing each step was reviewed earlier and is not repeated here.

First, for the full data set, with original (raw) data and separately with ipsatized data, we applied PA, a method that recommends a specific number of factors, to determine a conventional ceiling on the "numbers of factors" to consider. (For models above the conventional ceiling set by PA, see the sixth step below.)

Second, beginning with that "conventional ceiling" number of factors, we ran varimax, equamax, and oblimin for various numbers of factors. Previous work indicates that the latter two methods work well for identifying high-dimensionality structures, but varimax was also included because it has been the typical rotation method in studies of personality descriptors from the natural language. (Note: varimax and equamax produce orthogonal [uncorrelated] factors, oblimin allows for oblique [correlated] factors.) One prominent method of oblique rotation—promax—is intentionally omitted from the analyses because it is merely a variant of varimax that relaxes the orthogonality constraint and thus would tend to converge extremely highly (be redundant) with varimax.

Third, we discarded solutions that have any insufficiently sized factors by a preset "sufficient size" minimum of two salient terms with a loading of at least .40 in magnitude or alternatively two with at least .35 in absolute magnitude that also included at least one salient term with a loading of at least .45 in absolute magnitude. The essential criterion was that there should be at least two salient terms whose loadings are approximately .40 or higher.

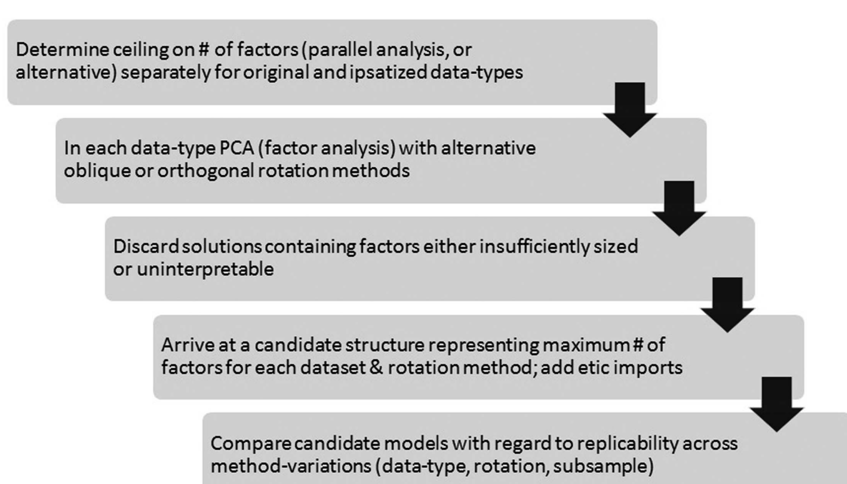
Fourth, for the solutions that remained (had all factors of sufficient size), we determined the maximum number of interpretable factors. Specifically, any solutions that have any factors that were judged by *neither* the second nor the fourth author as "possible to interpret substantively" were eliminated. (Note: Figure 1 consolidates this and the previous step into one stage.)

Fifth, for each of the remaining solutions—the maximum number of factors (which are sufficiently sized and interpretable) within each data set is set out as a candidate structure, to be compared with other alternative models at Step 7.

Sixth, because of potential benefits in terms of comprehensiveness and predictive capacity, we sought even higher dimensionality structures. Steps 1–5 rely on the assumption that PA provides an authoritative ceiling on the number of factors in a model. These steps were repeated without the same strict ceiling. Starting with a number of factors that PA recommends (let that be denoted as P), for both ipsatized and original data, we ran equamax solutions for $P + 1$

Figure 1

Essential Stages in the Analysis of High-Dimensional Personality-Attribute Structure



Note. PCA = principal component analysis.

factors and discarded the solution if there were any factors of insufficient size and then repeated that procedure for $P + 2$ factors, $P + 3$ factors, $P + 4$ factors, and $P + 5$ factors, ending the sequence of ever-increasing factor numbers only when at least five successive solutions each had at least one factor of insufficient size (at this point, the upward search was abandoned as insufficiently promising of useful results). We repeated the same procedure for oblimin solutions. Then, we examined solutions in which all factors were sufficiently sized (i.e., admissible solutions) starting with the one (in each of the four rotation-by-data type tracks, raw oblimin, ips. oblimin, etc.) that had the very largest number of factors and identified (by the procedure in Step 4) any factors that were interpretable. If any factor in a solution was found uninterpretable, that solution was discarded, and the admissible solution with the next highest number of factors is examined; this sequence continued until that admissible solution that had the very highest number of factors with all factors interpretable was identified. (Note: This could be at or below the number recommended by PA but was allowed to be higher than the recommended number by PA.) Any such models identified were retained as an additional candidate structure, alongside those models identified at Step 5.

To the extent that the procedures in the steps above did not examine solutions at the one- to six-factor level (emphasized in prior research), we planned to add these, with comparison on the same robustness indices. The one-factor level was indexed by the first unrotated principal component, and consistency with the previous literature would dictate that the two- to six-factor level be indexed by varimax-rotated components. These structures were examined primarily for comparison with high-dimensionality models (and with previous-study structures of one to six factors) with respect to robustness.

Seventh, in the final stage (as shown in Figure 1), all candidate models identified in Steps 2–6 were then compared across data sets and method variations, so as to sort out which number of factors has relatively better and worse convergence across methods (i.e., use of ipsatized vs. original/raw data, of orthogonal vs. oblique rotation, also more vs. less educated subsamples). Presumably, just one of the 10 emic-maximum candidate structures (i.e., the one from varimax, from equamax, and from oblimin for original data, and the same three for ipsatized data, and as many as four additional models identified at Step 6 in search for an emic maximum beyond the PA convention) would stand out as the most robust and the best emic-maximum model. “Converge across methods” as described here could also be described in terms of (lack of) sensitivity to method variation—alternative methods producing similar results—but we will use the abbreviated label “robustness” to refer to the same.

Steps 1 through 7 are all oriented toward identifying the optimal “emic-maximum” model from these data. The last step pertains to various “etic” models also used in this study’s comparisons. In this last step, we compared the obtained results with structural models from previous lexical studies. Appendix B from Saucier, Thalmayer, and Bel-Bahar’s (2014) study of person descriptors in 12 languages was used as a definitive set of reference terms for previous structures (i.e., structures having one to six factors). If a Persian term’s translation into English (combined translations by two sources completed prior to data analysis) appeared on the list of terms for a factor within a structure, it was included in the aggregate used to score that previous factor; the only exception to this simple rule was that terms that could be associated with any pair of factors within a structure were eliminated as markers/indicators for either of that pair, eliminating redundancy.

A few clarifications on specific methods employed in these preplanned analyses may be helpful. The orthogonal versus oblique comparisons were varimax versus oblimin for two- to six-factor solutions and for any varimax model. For all other comparisons, these were equamax versus oblimin comparisons, on the basis that as the number of factors becomes large, these two methods have been observed (by, e.g., Saucier & Iurino, 2020) to be able to produce more interpretable factors of sufficient size than can be obtained with varimax. As for the method for examining robustness, these necessarily varied according to the comparison: An ipsatized-original data comparison can utilize canonical-correlation analysis, which is an optimal method because it indexes variance accounted for regardless of arbitrary rotations of factor axes. The orthogonal–oblique comparisons needed to utilize the comparison of factor scores (the next most optimal method) because different rotations of the same unrotated factors produce singularity and so cannot be compared in canonical-correlation analysis. Comparison of structures from educated versus less educated subsamples (because these are the same variables but different cases) needed to rely on Tucker coefficients of factor congruence. In this study, models represented via Step 8 are examined only for comparison with high-dimensionality models (and with previous structures of one to six factors) to get at etic (cross-cultural) replicability.

Analyses were run in R wherever feasible and with SPSS (Version 27) where suitable R-package routines could not be found.

Results

PA indicated 20 factors for original (raw) data and 29 factors for ipsatized data. For the initial round of high-dimensionality analyses, these were set as ceiling for the number of factors, and we examined solutions with successively decreasing numbers of factors until identifying a solution with all factors being both sufficiently sized and interpretable.

Six differing structures meeting the criteria, and falling under the PA ceiling, were identified. Based on the original data, these were structures of eight varimax factors, 19 equamax factors, and 20 oblimin factors. Based on ipsatized data, these were structures of 12 varimax factors, 27 equamax factors, and 27 oblimin factors. For each “lane” of analysis (crossing data type with rotation method), no structure with more factors than these met the criteria of sufficient size and interpretability for all component factors. As anticipated, these intermediate steps in analysis led to six candidate structures to be compared. As in all previous studies using the current research approach, reliance on varimax rotation was the lane that led to the identification of the candidate structures with the fewest factors.

To identify any structures that might be even more comprehensive than these six, we sought structures of even higher dimensionality by taking PA recommendations as a floor rather than a ceiling on the number of factors. Proceeding upward, we were able to identify structures with as many as 50 factors—where all factors were sufficiently sized and interpretable. The specific candidate structures identified by this approach involved 50 equamax and 42 oblimin factors from original (raw) data, as well as 42 equamax and

30 oblimin factors from ipsatized data.[4] Adding these four candidates to the previous six, we had 10 high-dimensionality candidate structures to compare. These involved a considerable range in the number of factors from eight up to 50.

The 10 candidate structures identified by the procedures above all involved more than six factors. Therefore, as a supplement, to compare the robustness of few- with many-factor models, we went on to identify structures of one to six factors, emphasizing varimax rotation (except for the necessarily unrotated one-factor structure) and ipsatized data, so as to be most consistent with the bulk of previous studies.

The above results identify a set of 16 promising models, in total, to be compared. How do they compare with respect to robustness? Table 1 depicts the various robustness coefficients for structures of one up to 50 factors, including the 10 high-dimensionality candidate structures as well as the comparisons for one to six factors. To simplify take-home inferences from these many coefficients, they are presented alternatively in the form of a multiple-line graph in Figure 2. Figure 3 recasts the mean robustness values (from Table 1 and Figure 2) by comparing them with the percentage of variance accounted for by the structure, where the ideal—as indicated in the upper right of the figure—would be high comprehensiveness and informativeness (maximum percentage of variance) plus high robustness; as can be seen, the one-, eight-, 20-, and 42-factor models come closest to the ideal. Referring to the table and the figures reveals an interesting picture, although somewhat complex.

If one relied only upon one robustness index, the conclusions would differ according to the index of choice. If one relied only on replicability across more versus less educated subsamples, then one, two, five, and eight-factor structures would be favored; however, this is the robustness index most biased by limited sample size because it involves comparing two nearly half-sized subsamples. If one relied instead only on replicability across rotational method, structures of one to six factors (and perhaps one with 12 factors) would be favored. However, if one relied only on replicability between original and ipsatized data, structures of 12–50 factors are impressively more replicable than structures of two to eight factors; with an increasing number of factors, there tends to be greater resemblance between original and ipsatized data results The choice to ipsatize or use the original data, in other words, was of little consequence for the high-dimensionality structures but yielded more divergent results with fewer factors retained. Considerable significance adheres to the original versus ipsatized data contrast because lexical studies have conventionally employed ipsatized data as it reduces the impact of response biases, but downstream from there, personality assessment rarely ipsatizes, assuming one generalizes to the other; accordingly, agreement between these two types of data is highly desirable and relevant.

Combining the three indices into a mean (as in the heavy solid line on the graph), one can clearly conclude that the one-factor structure is the most robust in absolute terms. However, the fewer factors, the less comprehensiveness and (in general) predictive capacity, so it might be preferable to favor a more comprehensive (many-factored) structure over a very parsimonious one even if the robustness was slightly lower. Proceeding from the high peak at one factor to the slightly lower peak at eight factors, one loses some robustness but gains a great deal of comprehensiveness, yielding a trade-off. Similarly as one proceeds from the quite-high peak at eight factors to the somewhat high peak at 20 factors, one has the same kind of trade-off. Then again, as one proceeds from the somewhat high peak at 20 factors to the slightly high peak at 42 factors, the same trade-off arises.

The best resolution of the trade-offs between absolutely maximizing robustness and absolutely maximizing comprehensiveness would be to conceive of personality structure in terms of several levels or tiers. Figures 2 and 3 indicate optimal levels, for Persian adjectival descriptors, to be at one, eight, 20, and 42 dimensions. Compared with the other candidate structures, these structures achieve a better ratio of robustness to comprehensiveness (and likely predictiveness). Figure 4 maps the relation of these four structures, including the correlations between factor scores for the constituent factors wherein those correlations exceed .40 in magnitude. Figure 4 is divided into a top and bottom part to enhance its intelligibility to the reader.

The foregoing results point to four alternative structures that, taking into account varying criteria, have roughly equivalent levels of support. To what extent are relations among these structures hierarchical, and how highly intercorrelated are the factors within each structure?

If relations among these levels were a pure hierarchy—as one would encounter in contemporary "facet" models of personality structure—then all factors at the more specific levels would be well-linked to the factors at the more broad levels. However, only three of the factors in the set of eight have a >.40 correlation with the higher level one-factor structure; if the criterion was a .50-magnitude correlation, only one of the eight would qualify.[5] Nineteen of the factors in the set of 20 have a >.40 magnitude correlation with the higher level eight-factor structure, which does indicate considerable hierarchy at a loose or lenient level; with a stricter criterion of a .50-magnitude correlation, only 13 of the 20 would qualify as hierarchically related. Of the factors in the set of 42 factors, 33 have a >.40 correlation with a component of the higher level 20-factor structure, but only 25 of the 42 have a correlation >.50 in magnitude, so the hierarchy is again quite incomplete, especially if one uses the stricter (.50) criterion. In sum, although the differing levels of robust structure are at least somewhat hierarchical in nature, a substantial fraction of factors in the higher dimensionality models (20 and especially 42 factors) does not give strong evidence of hierarchical

[4] A reviewer wondered if using the MAP criterion would have led to a differing outcome. MAP was not used here due to concerns raised by Saucier and Iurino (2020) about its odd results in cases of very large numbers of single items. When we ran MAP on a post hoc basis for comparison, we found that differing R packages for its computations yielded differing results (numbers of factors as a ceiling), but all fell in the intermediate range between what our pair of procedures yielded, always recommending 39–52 factors. The original MAP criterion of Velicer (1976) would have recommended 39 or 40 factors for original data, and thus, had we relied on it as a definitive ceiling, we would have missed the very useful 42-factor structure.

[5] One could object that a contrast of original versus ipsatized data structures is the source of the low numbers because the one-factor candidate model is based on ipsatized data and those with eight, 20, and 42 are based on original data. However, comparing one- and eight-factor structures based (in both cases) on original data led to a similar outcome: Only two of eight met either the >.50 or the >.40 threshold. Of note, and consistent with the pattern of merely partial hierarchy, 28 of the 42 fine-grained factors had a correlation >.40 with one of the eight factors, but only 13 of 42 had a correlation >.50 in magnitude. The analyses in this paragraph are post hoc and not in the preregistration.

Table 1
Robustness Indices for One to Six Factors and 10 High-Dimensionality Candidate Structures in Persian Lexical Data

Candidate model	Sum shared variance proportion ipsatized vs. original	Average squared orthogonal–oblique best match correlation	Educated vs. less educated mean best match congruence	Mean robustness
One factor (unrotated)	0.82	1.00	0.95	**0.92**
Two factors (varimax)	0.57	0.99	0.94	**0.83**
Three factors (varimax)	0.65	0.99	0.64	**0.76**
Four factors (varimax)	0.72	0.98	0.88	**0.86**
Five factors (varimax)	0.7	0.97	0.90	**0.86**
Six factors (varimax)	0.74	0.96	0.80	**0.84**
Eight factors (varimax)*	0.80	0.90	0.90	**0.86**
12 factors (varimax)	0.84	0.94	0.65	**0.81**
19 factors (equamax)*	0.84	0.89	0.75	**0.83**
20 factors (oblimin)*	0.86	0.82	0.81	**0.83**
27 factors (oblimin)	0.86	0.73	0.67	**0.75**
27 factors (equamax)	0.85	0.73	0.60	**0.73**
30 factors (oblimin)	0.85	0.84	0.67	**0.79**
42 factors (oblimin)*	0.84	0.74	0.78	**0.79**
42 factors (equamax)	0.81	0.87	0.56	**0.74**
50 factors (equamax)*	0.82	0.67	0.63	**0.71**

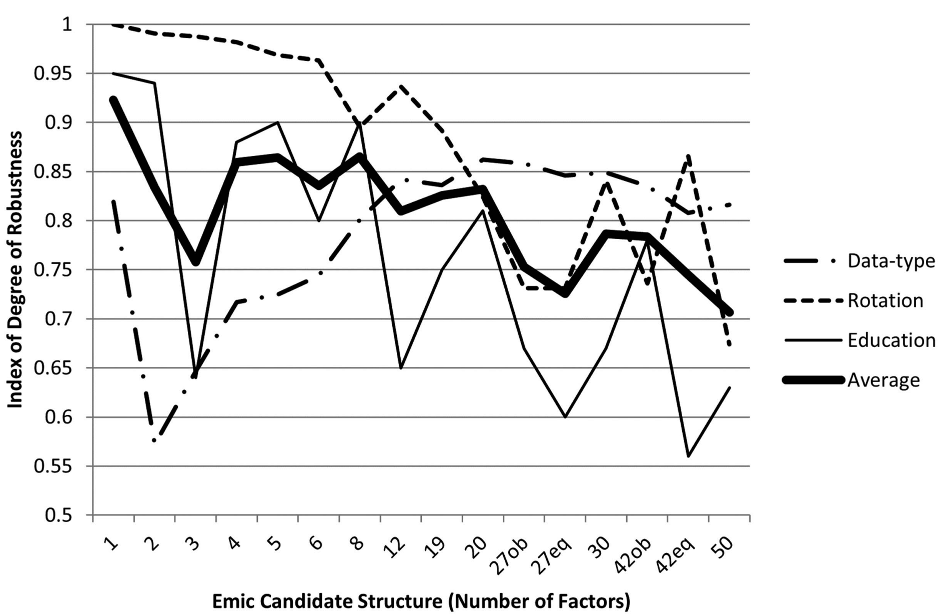
Note. $N = 767$. Ipsatized data, except that the models marked with * are derived from the original data. Values presented in bold are the mean.

relation with more parsimonious structures. There appears to be some hierarchy but also much that gets around the hierarchy.

As has been observed among English nouns by Vader and Saucier (2025), findings here indicate rather robust models above (in number) the limit set by PA. The earliest studies using the high-dimensionality approach (Saucier et al., 2020; Saucier & Iurino, 2020) relied entirely on PA as an authoritative limit. The present results indicate that PA can lead one to the lower part of high-dimensionality structure (e.g., 20 factors, here, as in Saucier & Iurino's, 2020, study), but one can identify a stratum (e.g., 42 factors here) above what PA would seem to allow.

A common (but unwarranted) assumption about high-dimensionality structures is that with so many factors (e.g., 20 or 42), the factors must be substantially correlated with one another. The assumption is based on the familiar example of so-called "facet" structures of personality, in which the more specific components are quite intentionally strongly linked to broad-level components and (thus) to each other. However—as previously demonstrated by Saucier and

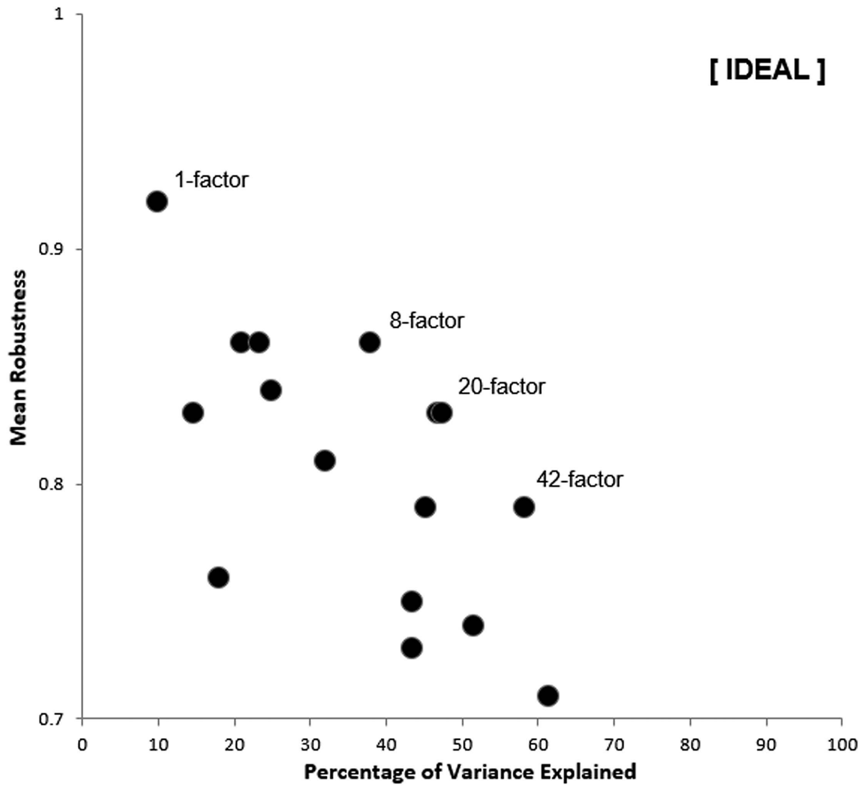
Figure 2
Robustness of Candidate Structures Across Three Kinds of Method Variation



Note. To disambiguate pairs of structures with the same number of factors: 27ob = 27 oblimin factors; 27eq = 27 equamax factors; 42ob = 42 oblimin factors; 42eq = 42 equamax factors. The range on the vertical axis is truncated to run from 0.5 to 1.0.

Figure 3

Percentage of Variance for Each Structure Plotted Against Its Mean Robustness



Note. Mean robustness values from Table 1 are plotted against the percentage of variance explained by the structure, with those for the best performing structures labeled. The range on the vertical axis is truncated to run from 0.7 to 1.0. The ideal values would be in the upper right (1.0 mean robustness, 100% of variance).

Iurino (2020) as well as Vader and Saucier (2025)—when one uses an oblique (conventional oblimin) rotation allowing the lexical-personality factors to correlate, the factor intercorrelations are quite modest. For the 20-factor emic-Persian structure, the highest factor intercorrelation was magnitude .30, and only 11 of the 190 factor intercorrelations exceeded magnitude .25. For the 42-factor structure, the highest factor intercorrelation was magnitude .31, and only six of the 861 factor intercorrelations exceeded magnitude .25. This (again) goes against the common assumption that there are only five or six relatively independent dimensions of personality.

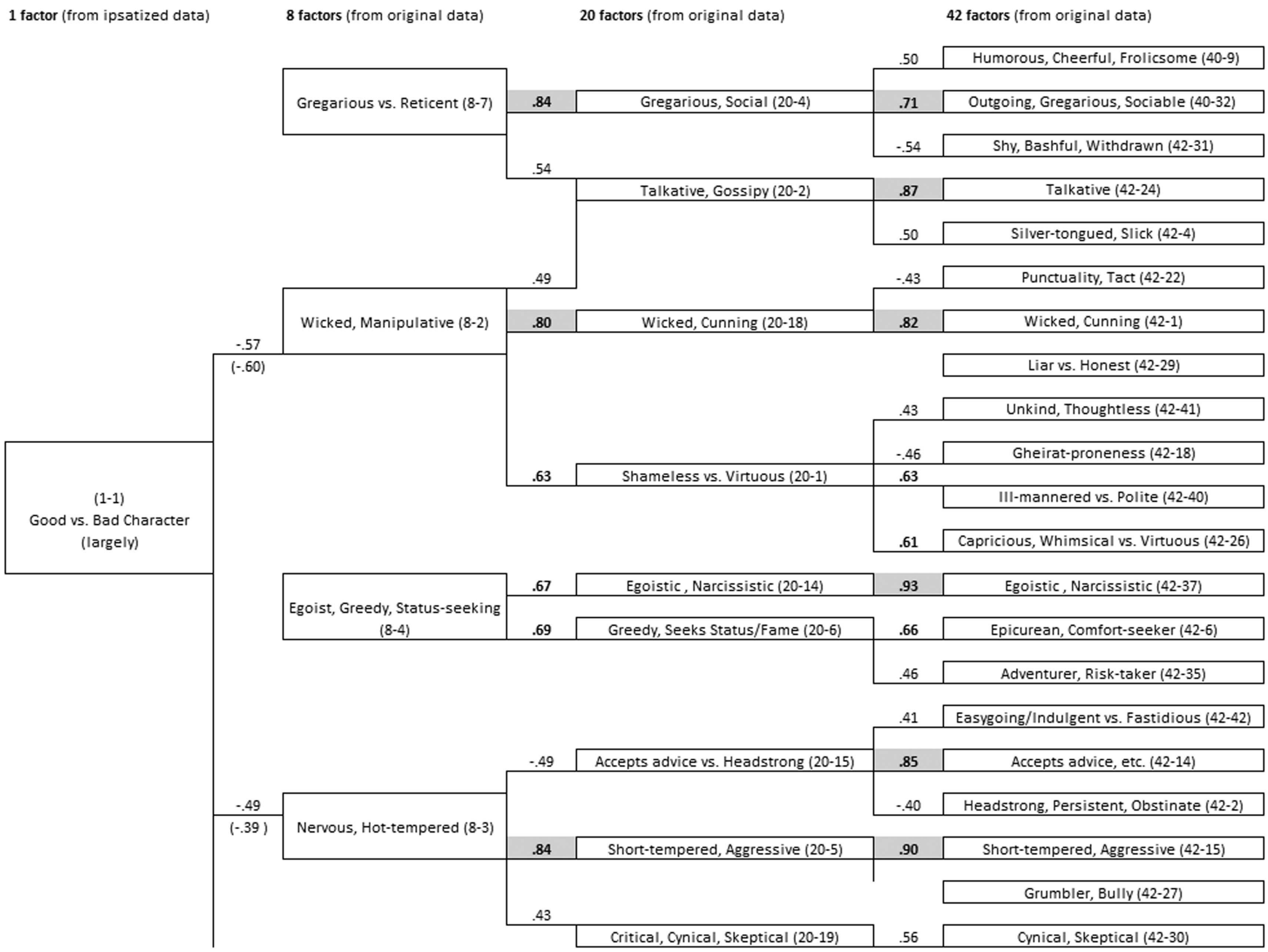
Therefore, we can conclude that—contrary to the assumptions of so-called facet models of personality—personality attributes can be held to include several dozen specific fine-grained constructs that are relatively independent of one another. Moreover, the constructs at the fine-grained level are not necessarily much related to those at the broad-factor level, so broad-factor-only representations are likely in large part to miss them, which brings us to the next issue: How do emic-Persian structures compare with etic (imposed) structures from previous research, such as the Big Five?

We compared imported (etic) models of one with six factors—based on reference markers from Saucier, Thalmayer, and Bel-Bahar (2014)—to emic-Persian structures having the same number of factors based on the present lexical-study data. The results are presented in Table 2 and graphed in Figure 5. The only truly impressive emic replication of an etic structure was for the Big Two (social self-regulation and dynamism; Saucier, Thalmayer, Payne, et al., 2014). Replication of imported three-, four-, and six-factor models was more modest, but with best match correlations averaging at or near .60 or higher in magnitude for these models; one might label these as at least a weak or partial replication. The imported Big Five structure had a weaker result: A principal culprit was the lack of an intellect or imagination (or "openness") factor. The emic factor least correlated with the other four of the Big Five had dishonesty content (e.g., wicked, manipulative, ruthless) appearing instead of intellect content. The limitation on the replication of the six-factor model was that (dis)honesty and humility content were split into two different orthogonal emic-Persian factors and neither of these was related much with etic-imported "openness."

The one-factor model had the highest robustness within the Persian data, yet it did not demonstrate good etic replicability. However, this analysis may underestimate replicability because the only etic-marker terms available in the present data tended to involve agreeableness (kind, friendly, good natured) and not broader more abstract content (e.g., good, bad, beautiful), whereas the emic-Persian one-factor solution though still had very evaluative and heterogeneous content. Therefore, the modest etic replicability may

Figure 4

Depiction of Content in and Relations Among Structures of One, Eight, 20, and 42 Factors



(*figure continues*)

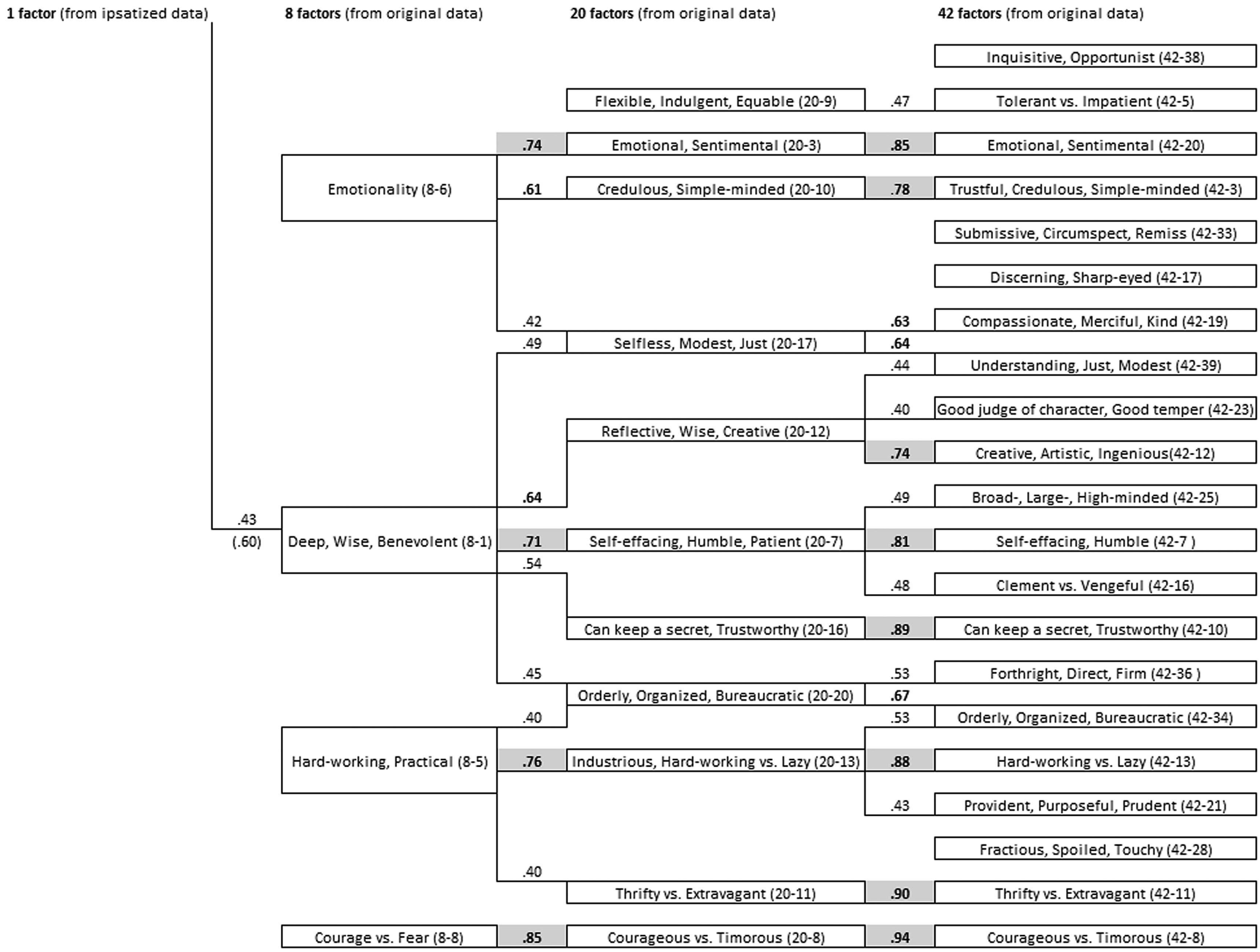
simply indicate that the emic-Persian "Big One" evaluation factor is much broader than just core-agreeableness content.

Saucier, Thalmayer, and Bel-Behar (2014) did not provide (and could not have provided) marker terms for the 20-factor solution identified in a later study by Saucier and Iurino (2020) and that 20-factor structure as yet has no standard set of marker terms. Informally, though, if one compares the labels (for 20 factors, i.e., "Lex-20") offered in Table 11 of the latter article, one can make two observations. First, four of the Lex-20—prejudice, sophistication, conventionality, and reflectiveness—would be quite unlikely to appear because the content of these types had been eliminated at an earlier stage of classification in Persian. Second, a majority—as many as 12—of the 16 remaining Lex-20 factors (from North American English data) seem to reappear content-wise in the Persian 20-factor structure. This informal comparison does serve to suggest that high-dimensionality structures in one language versus another might be expected to have (a) a good deal of overlap in their content, with the overlap involving more than just five or six factors, but also (b) areas of nonoverlap, with (c) the degree of nonoverlap exacerbated where there is divergence in content inclusion (i.e., variable selection) from one data set to another. The same issues arise, of course, in comparing low-dimensionality factors found in lexical studies of various languages, there being overlap (e.g., one always sees some broader or narrower dimension related to agreeableness), nonoverlap (e.g., regarding intellect/imagination/openness in particular), and differences arising because of variable-selection exclusions (e.g., Saucier, 2019). It is not clear yet whether the degree of nonoverlap (beyond that caused by variable-selection exclusions) is *proportionally* greater for high- than for low-dimensionality structures of personality. More formal and rigorous comparisons, involving more than just two languages, are needed in the future to resolve this issue.

Discussion

Several main points can be gleaned from the results. A well-accepted indicator of the number of viable factors in a data set, PA, indicates well more than five or six factors among Persian lexical

Figure 4 (*continued*)



Note. Horizontal lines link those factors from differing structures having factor-score correlations of .40 or above, in absolute value. Coefficients with a magnitude over .60 are presented in bold, and those over .70 are shaded. The coefficients in parentheses link the eight-factor structure with the one-factor structure from the original/raw data.

personality descriptors. However, herein (as in Vader & Saucier, 2025) reasonable and relatively robust structures with far more factors can be identified. There are trade-offs between parsimony and absolute robustness (which favor few-factor models) and comprehensiveness and predictive capacity (which favor high-dimensionality models). Balancing these trade-offs, these Persian-language results point uniquely toward four tiers of emic personality structure in the lexicon, involving one, eight, 20, and 42 factors. The 42-factor level in particular includes some apparently culture-specific constructs (discussed further below).

Other features in our results are more likely to be shared with other languages and cultural contexts in future studies. While there are some hierarchical relations between these tiers of structure, much of the relation among these tiers is nonhierarchical, with the more complex and differentiated tiers pulling in a good deal of content not well accounted for at the simpler tiers. The complex/differentiated tiers involve (perhaps surprisingly to some readers) factors that have at most quite modest levels of intercorrelation—unlike what is expected and found among so-called "facet" models of personality. Examining the relative degree of replication of various few-factor models long emphasized in lexical studies, the Big Two (social self-regulation and dynamism) were distinctly the structure most impressively replicated; structures of five and six factors had a considerably lower level of replication in these data, although the etic six-factor model fared better than a five-factor (Big Five) model.

Why so Many Dimensions of Personality?

This is not the first study to find as many as 20 factors among personality descriptors (Saucier et al., 2020; Saucier & Iurino, 2020; Vader & Saucier, 2025) and is far from the first to find a two-factor structure to be highly robust (Saucier, Thalmayer, Payne, et al., 2014). However, the finding of a highly differentiated tier of as many as 42 factors is novel. Given the rather large body of previous lexical studies, why this novel result? One possibility involves the higher-than-usual typical level of educational attainment in this

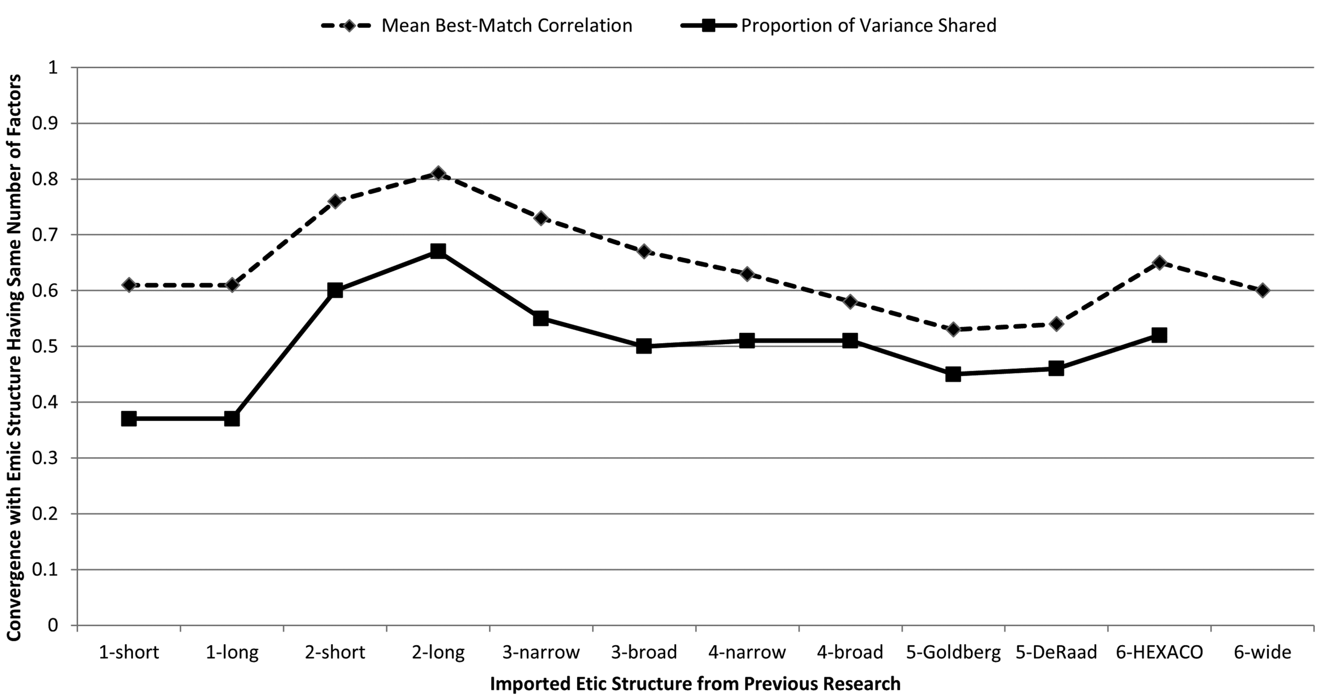
Table 2
Degree of Replication for Etic-Imposed Structures of One to Six Factors Based on Marker Items Set Out in a 2014 Study

Etic-imposed structure	1st	2nd	3rd	4th	5th	6th	*M*	SPSV
Big One—shorter markers	**.61**						**0.61**	**.37**
Big One—longer markers	**.61**						**0.61**	**.37**
Big Two—shorter markers	**.83**	**.70**					**0.76**	**.60**
Big Two—longer markers	**.85**	**.77**					**0.81**	**.67**
Big Three—narrow interpretations	**.75**	**.73**	**.72**				**0.73**	**.55**
Big Three—broad interpretations	**.69**	**.67**	**.65**				**0.67**	**.50**
Big Four—narrow interpretations	**.75**	**.72**	**.70**	.36			**0.63**	**.51**
Big Four—broad interpretations	**.77**	**.72**	**.67**	.18			0.58	**.51**
Big Five—Goldberg markers	**.71**	**.62**	.59	.52	.20		0.53	**.45**
Big Five—De Raad et al.'s markers	**.73**	**.70**	**.60**	.48	.18		0.54	**.46**
Big Six—HEXACO-like	**.82**	**.80**	**.77**	**.72**	**.63**	.17	**0.65**	**.52**
Big Six—wide variable selection	**.80**	**.74**	**.66**	.54	.26	—	**0.60**	—

Note. $N = 767$. — = because insufficient marker content was available in the data's variables to represent one of the wide Big Six factors, this correlation could not be computed. Correlations over .60 and SPSVs over .36 (i.e., $.60^2$) are presented in bold. SPSV = summed proportion of shared variance calculated via canonical-correlation analysis. HEXACO = humility, emotionality, eXtraversion, agreeableness, conscientiousness, openness.

sample; over half the sample reported having graduate degrees. Higher education might lead not only to larger vocabularies but also more sensitivity to the meanings and nuances of linguistic terms, and possibly more self-knowledge, which could all aid and abet the emergence of a more differentiated structure. Moreover, it is possible that greater education tends to facilitate more individualism, and alongside the valuing of uniqueness, individual differences might increase. Some previous studies (e.g., Rammstedt et al., 2010; Toomela, 2003) have indicated that personality-attribute structure tends to be less differentiated in less educated samples, and we may be observing in the present study the other (high-educated) end of the spectrum. One can certainly debate about the relative utility of sampling a representative set of laypersons from a population (with more fidelity to what common-denominator perceptions are, even if they tend to be simplifying) versus sampling those who have more expertise in language use (which may lead to a structure with

Figure 5
Graph of Mean Degree of Replication for Etic-Imposed Structures of One to Six Factors



Note. "Mean best match correlation" represents the average correlation between pairs of matched factors (each pair including an etic-imposed factor and its closest counterpart in the emic structure) wherein each factor appears in only one pair and is accomplished in such a way that the average is maximized. "Proportion of variance shared" is derived from canonical-correlation analysis of the etic and the emic set for that specified number of factors. HEXACO = humility, emotionality, eXtraversion, agreeableness, conscientiousness, openness.

more insight-generating potential). It may be that the same lines of variation exist latently among the less educated, although the less educated are themselves less prone to recognize them. Block (1995, p. 197) quite anticipated this issue, questioning whether undergraduates have sufficiently "reached that mature stage of development where they can function as criterion judges for the ultimate selection of personality discriminanda and serve as expert evaluators and describers of personality" and "be accepted as the standard for scientific study within the field."

However, another explanation would be rooted in statistical rather than sampling methodology. Previous studies have either relied upon inspection of the large-scale pattern in unrotated-factor scree plots (as in studies emphasizing five or six factors) or relied upon PA to set an upper limit. As shown in another recent study with type-nouns rather than adjectives (Vader & Saucier, 2025), it is fruitful to search for a coherent structure well *above* the number of factors that PA recommends. As noted by Vader and Saucier (2025), meaningful *rotated* factors can arise that are no larger than some *unrotated* factors from randomly generated data, and so the reliance of PA on examining unrotated factors may produce a blind spot. Accordingly, the structure PA identifies tends to be at an intermediate and not at the most fine-grained level of personality-attribute structure.

Hierarchical Tiers of Personality-Attribute Structure

The four tiers of structure identified here are meaningfully different in the kind of lens they provide for viewing personality. The broadest tier (with one or two dimensions) is advantageously parsimonious, easy to remember, and easy to link to theory (see, e.g., Digman, 1997) though its factors can be critiqued as overly abstract and evaluative, with overly heterogeneous factors. The next tier, the one at which Big Five and humility, emotionality, eXtraversion, agreeableness, conscientiousness, openness structures are found, retains some parsimony and may have memorability advantages because their number of factors is in the vicinity of the high end of working-memory capacity (Cowan, 2010), but factors at this tier are characteristically all collages of differentiable subcomponents; although these collages may not be arbitrary, they can certainly be culture specific in their formation pattern, and factors at this few-factor level are also characteristically rather evaluative. The third tier (e.g., with 20 factors) essentially contains the largest of the more granular and descriptive dimensions; it tends to break up the collages, so that factors have more singular interpretation, and the factors are less characteristically prone to have an evaluative cast, but which factors emerge as the largest of these singular-specific-descriptive dimensions—big enough to pass the PA threshold test—may itself be partly culture specific. The fourth tier factors (e.g., 42 dimensions in this study) have perhaps an uncomfortable level of complexity and may occasionally identify aspects of personality so subtle and specific that many people cannot easily recognize them; however, they are singularly interpretable, provide the most comprehensive (and potentially predictive) set, and may be the best place to look for culture-specific personality concepts.

Therefore, how many dimensions/factors of personality are there? It depends on how broad and abstract and strongly or largely evaluative—or how specific and descriptive—one wishes to be and whether one (as one rationally should) emphasizes cross-cultural comparability. While there is substantial evidence that the broadest (and most evaluative) tier is the most cross-culturally universal (De Raad et al., 2010; Osgood et al., 1975; Saucier, Thalmayer, & Bel-Bahar, 2014; Saucier, Thalmayer, Payne, et al., 2014), it is not yet clear whether or how much the next tier (the one that would include humility, emotionality, eXtraversion, agreeableness, conscientiousness, openness Big Five and models) is more universal than the more differentiated tiers. From one fifth to two fifths (20%–40%) of the Big Five model has been difficult to replicate universally in lexical studies (De Raad et al., 2010, 2014), and we do not yet know whether comparable percentages of factors at the more differentiated descriptive tiers will also be difficult to replicate.

Nonuniversal and Culture-Specific Aspects of Personality

The proneness of the broad-factor level to generate nonuniversal aspects can be illustrated in a large number of lexical studies and can be illustrated in several respects in this study. As described in the Results section, Persian six-factor structures resembled humility, emotionality, eXtraversion, agreeableness, conscientiousness, openness factors more than Persian five-factor structures resembled the Big Five, but the six-factor correspondences are limited particularly by (a) the absence of an openness factor in the Persian data and (b) the six- (as well as the eight-) factor structure splitting the honesty–humility collage into separate deviousness (low honesty) and egoism (low humility) factors. Such a differentiation implies that one could observe humility without honesty. There is a culture-specific logic for why this might arise in the Persian context. Unlike the American context, in the Persian context, humility is a very highly valued and emphasized trait, generating expectations so strong that one might observe "performative humility"—dramatic demonstrations of humility that may sometimes be fake or inauthentic. (Analogously, in a cultural context wherein joy and cheerfulness are highly valued and emphasized, one might sometimes observe performative demonstrations of such that are excessive or faked to satisfy social expectations, expectations such as "one should be smiling in all photographs.") The point is that broad factor collages rest on the assumption that constituents of the collage tend to be inseparable, but aspects of cultural contexts can make these constituents more versus less separable.

The absence of a clear *broad* openness-related factor here might be attributed to any one (or a combination of) the following: (a) the lesser tendency for such a factor to emerge in non-Western contexts (e.g., Cheung et al., 2001; Zeinoun et al., 2018), (b) judges removing openness-related terms by classifying them not as traits but within the exclusion category of abilities or talent, and (c) restricted range in openness-related variables in a relatively highly educated sample, which would tend to lower intercorrelations among such variables.

We tend to believe that at least three of the factors in the highly differentiated 42-factor solution will turn out justifiably to be considered culture specific. Before discussing them, it is useful to clarify on what bases a psychological construct might be judged as culture specific. The first criterion is difficulty of translation into most or all other languages, that is, a single-word concept that is the best label for the construct in the one language requires a lengthy explanation (beyond just a phrase) in other languages. The second criterion is a certain culture *boundedness*, that is, the concept is awkward—and perhaps impossible—to use in other cultural-linguistic contexts because the use of the concept requires certain assumptions that do not hold very widely across cultural contexts;

thus, the concept will tend to be easily misunderstood outside its home context. The third criterion is that the concept is relatively important and frequently referenced in the home context; because all variables used in this study are frequently used concepts, all factors examined in this study would fulfill this third criterion.

One such culture-specific concept of personality is *gheirat* proneness. *Gheirat* is a moral emotional experience ubiquitous in Iran and many other Muslim cultures, which is often elicited in response to a heterogeneous set of relational violations such as perceived harm or insult to the extended self or romantic betrayals involving one's partner or kin (Atari et al., 2020; Bakhtiar, 2015; Razavi et al., 2023). *Gheirat* proneness meets all the criteria discussed above. First, there is no direct translation for *gheirat*. It is possible to use contiguous concepts (e.g., romantic jealousy, anger, honor, and intrasexual rivalry) to communicate some *gheirat* experiences. However, using these concepts would not capture the complete meaning of *gheirat* proneness and experiences associated with it (Razavi et al., 2023). In other words, to an Iranian person, describing someone as *gheirat* prone (i.e., *gheiratmand*, *bagheirat*, and *gheirati*) provides information about their character that is not fully captured by terms such as jealous, angry, or honorable. Second, some of the values and situational appraisals that give rise to the experience of *gheirat* are culturally bounded and might not hold widely beyond Muslim/Iranian cultural contexts. For example, some of the violations that can elicit *gheirat* in an Iranian person are dependent on adherence to Islamic cultural norms or traditions such as *hijab* or *mahramiat* (Moradinasab, 2021) and might not be relatable to the lived experiences of people who do not subscribe to (or are not closely familiar with) such norms.

Another likely culture-specific concept is *azadmanesh*. The two highest loading trait adjectives for this dimension (i.e., *azadeh* and *azadmanesh*) are strongly valued character descriptors for a person who has high moral standards, does not give up their principles easily (e.g., as a result of social pressures or worldly temptations), and is willing to make sacrifices to resist oppression (Moradi Ghiasabadi, 2016; Mostafavi, 2022). One translation for *azadmanesh* might be "high minded" or "large minded." However, this translation misses the connotative aspect of doing so as a nonconformist who is steadfast despite temptations and does not submit to social pressures or oppressive constraints.

A third possibility is the concept of *mosamehe-kar*, which includes trait adjectives translated as compromiser, submissive, and circumspect. Even though the notion of being cautious or circumspect is certainly not specific to the Persian context, the inclusion of other associated traits (i.e., the tendency to compromise or submit to authority) adds a subtle, but noteworthy, distinction to this dimension. These additional associations might have to do with the possibility that certain behaviors and characteristics become more relevant to social processes in a tight or authoritarian cultural context (Dimant et al., 2025). As a result, the extent to which individuals have a tendency to compromise, "hunker down," or submit to (or accept the will of) authority might receive heightened social (and, consequently, linguistic) attention, which will lead to the emergence of a distinct trait-adjective dimension. In other words, in certain contexts, more attention gets paid to who is (and who is not) regulating or suppressing their behavior so as not to stand out.

Although these three culture-specific dimensions here make their first appearance in a lexical study of personality, they may not be entirely specific to the Persian-language context. Future research can seek to identify whether similar constructs spontaneously arise in other contexts (e.g., other Middle Eastern ones).

Limitations and Future Directions

It could be argued that this study's employment of a classification scheme for person descriptors first developed for use with the German language (Angleitner et al., 1990) is an etic imposition that essentially forced a Western definition of personality traits onto the Persian context. Of course, this scheme has been used widely in studies of numerous languages (e.g., Zeinoun et al., 2018), and its use here facilitates comparability with those studies. Moreover, it was Persian (not German) judges who made the judgments. Nonetheless, it would be useful for future research in the Persian context to examine whether a wider inclusion of dispositional types affects the kind of structures identified in this study.

An important future direction in lexical research is identification of an optimal start-point criterion for identifying the most differentiated aspect of high-dimensionality attributes, at least in the personality domain. PA has shown utility for identifying a fairly complex, differentiated level well above the Big Few, but as illustrated here (and by Vader & Saucier, 2025), there are viable and rather robust structures above and beyond what PA indicates. Likely one can fall back on one of two heuristic rules long evident in the factor-analysis literature. One involves highlighting whatever number of factors accounts for 50% of the variance in correlation matrix (e.g., Merenda, 1997), and the other highlights whatever number of factors accounts for 60% of the variance (e.g., Hair et al., 2010). The present study used neither rule but rather used PA as a lower bound and worked upward—a thorough but perhaps unnecessarily time-consuming approach. It is worthwhile to consider how the use of these two rules might have led to differing results than those we obtained, thus how they would apply to the original (nonipsatized) structures like those we identified as most robust (at eight, 20, and 42 factors). Had we used the 50% rule, we would have selected 25 factors as a starting point. Had we used the 60% rule, we would have selected 47 factors as the starting point. In the original (nonipsatized) data, there was no structure with more than 42 factors wherein all factors were sufficiently sized and interpretable and wherein the structure was also reasonably robust across method variations. The 42-factor solution accounted for 58% of the variance. Thus, the 50% rule was undershooting the target, and the 60% rule was overshooting it. The target solution could have been reached by either starting at the 50% solution and going upward or starting at the 60% solution and going down; because it sets an exact ceiling on factor number, the 60% approach seems ultimately more efficient. Future research would do well to explore the comparative utility of the 50% and 60% rules, as a potentially indispensable supplement to PA in the goal of identifying robust high-dimensional personality structure.

It could be argued that the structures derived from the PA starting point are dispensable because more comprehensive and predictive structures can be identified by disregarding PA and using a 50% or 60% rule. On the other hand, in the long run, (a) the PA starting point may lead to more replicable structures than these other rules, or (b) turn out to be easy to measure in a wider variety of contexts, or (c) for some reason (e.g., the appearance of exactly 20 robust factors in very many languages) cross-cultural comparability will be enhanced by keeping the PA starting point within the method paradigm.

Therefore, we recommend that for the time being both PA and the 50% or 60% rule be employed side by side.

Beyond possible tweaks to inform methods for future studies of high-dimensionality structure, what are the practical implications of this study's findings? Personality scientists should be more cautious about overclaims regarding the universality of Big Five and six-factor models and their suitability for non-Western contexts and not make assumptions that specific features of personality are all subsumable under such broad-factor models. Moreover, they should be more alert to the possibility of truly culture-specific personality constructs, which seem to reveal themselves more clearly as one moves beyond broad-factor models. Finally, this study lends support to the value of assessing personality at the level of fine-grained constructs. As for which fine-grained level (e.g., eight or 20 or 42 dimensions) is optimal, specifically in the Persian-language context or in general, that should depend on future research comparing the utility of such alternatives.

As was true of early lexical-study results that supported the Big Five (e.g., Goldberg, 1990), the structures identified here could provide the template for personality assessments tailored specifically to Persian-language contexts. However, the production of an assessment instrument was not the aim of this study, and it would be helpful for future research to adjudicate better which fine-grained level of structure is the most optimal for future applications.

Conclusions

This study leads to several major substantive conclusions. First, the natural and spontaneous organization of personality attributes in the Persian context does not support the Big Five and six-factor models currently popular in the West. Depending on how differentiated one prefers to have one's personality constructs, it better supports structures of one, eight, 20, or 42 dimensions. While the four supported structures might be understood as four hierarchical tiers, the hierarchy is only partial because much content at more differentiated levels is unrelated to the broader levels. Finally, the most differentiated (42-factor) structure contains three likely culture-specific factors, as well as an interesting strong separation (at the broad level) between honesty and humility contents.

A currently commonplace assumption became established in recent decades that variation between individuals in their personality characteristics is best summarized in terms of five—or at most six—dimensions. The present application of the lexical approach to the Persian language refutes this claim. Beyond the benefit of greater comprehensiveness and likely greater predictive capacity than few-factor models, the inclusion in high-dimensionality tiers of both potentially culture-specific aspects of personality and more universal fine-grained ones makes these higher tiers of structure promisingly beneficial for understanding relations of culture and personality.

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Received December 8, 2023
Revision received February 12, 2025
Accepted March 10, 2025 ■